Types of Differentiated Services Code Point (DSCP)

Introduction

Differentiated Services Code Point (DSCP) is a field in the IP header used for packet classification in IP networks, forming the basis of Quality of Service (QoS). DSCP enables routers and switches to prioritize different types of traffic, ensuring efficient delivery of latency-sensitive applications like VoIP, video conferencing, and real-time gaming.

The DSCP field is six bits wide, allowing for 64 different codepoints, each corresponding to a specific forwarding behavior, known as Per-Hop Behavior (PHB). DSCP replaced the older IP precedence system, expanding the classification range and enabling more sophisticated traffic handling.

DSCP Value Ranges

DSCP values fall into three main categories:

1. **Default Forwarding (DF)**
   - **DSCP Value**: 000000 (Decimal 0)
   - **Use**: Best-effort traffic with no QoS guarantees.
2. **Class Selector PHBs (Backward-Compatible with IP Precedence)**
   - **DSCP Values**: 001000 (Decimal 8) to 111000 (Decimal 56), in steps of 8.
   - **Use**: Maintains compatibility with older IP Precedence-based systems.
   - **Examples**:
     - CS0: 000000 (Best Effort)
     - CS1: 001000 (Priority 1)
     - CS2: 010000 (Priority 2), etc.
3. **Assured Forwarding (AF)**

- **DSCP Values**: AFxxy format, where x = class (1–4), y = drop precedence (1–3)
- **Use**: Provides different levels of assurance based on drop precedence.
- **Examples**:
    - AF11: 001010 (Decimal 10)
    - AF21: 010010 (Decimal 18)
    - AF31: 011010 (Decimal 26)
    - AF41: 100010 (Decimal 34)

4. **Expedited Forwarding (EF)**
    - **DSCP Value**: 101110 (Decimal 46)
    - **Use**: Low-loss, low-latency, low-jitter, and assured bandwidth—ideal for real-time traffic like VoIP.
5. **Voice Admit (VA)**
    - **DSCP Value**: 101100 (Decimal 44)
    - **Use**: Admitted voice traffic that complies with admission control policies.
6. **Local/Experimental Use**
    - **DSCP Values**: 011111 (Decimal 31), 101111 (Decimal 47), 111111 (Decimal 63), etc.
    - **Use**: Reserved for experimental or local administrative usage.

Table of Common DSCP Values

Name	Binary	Decimal	Description
CS0	000000	0	Best effort
CS1	001000	8	Scavenger or low-priority traffic
AF11	001010	10	Class 1, Low Drop Precedence
AF12	001100	12	Class 1, Medium Drop
AF13	001110	14	Class 1, High Drop
AF21	010010	18	Class 2, Low Drop Precedence

Name	Binary	Decimal	Description
AF31	011010	26	Class 3, Low Drop Precedence
AF41	100010	34	Class 4, Low Drop Precedence
EF	101110	46	Expedited Forwarding (VoIP)
VA	101100	44	Voice Admit (Call Signaling)
CS7	111000	56	Network Control Traffic

Applications of DSCP

- **VoIP and Real-Time Applications**: Use EF to ensure minimal delay.
- **Streaming Media**: Use AF classes to balance bandwidth and drop sensitivity.
- **Background Data (e.g., backups)**: Use CS1 or scavenger classes.
- **Critical Control Traffic**: Use CS6 or CS7 for routing and network control protocols.

DSCP in Practice

In enterprise and service provider networks, DSCP values are used in QoS policies:

- **Marking**: Assign DSCP values at the network edge.
- **Classification**: Identify packet types based on DSCP.
- **Queuing**: Assign traffic to queues with different priorities.
- **Policing/Shaping**: Control bandwidth use and enforce limits.

Example (Cisco QoS Configuration)

```
class-map match-any VOICE
 match dscp ef
!
policy-map QOS_POLICY
 class VOICE
  priority percent 30
```

This configuration prioritizes EF-marked traffic with 30% bandwidth allocation.

Conclusion

DSCP provides a scalable and efficient mechanism for classifying and managing IP traffic. Its integration into QoS policies allows networks to provide differentiated levels of service based on application requirements, ensuring robust performance for latency-sensitive and mission-critical traffic.

References

- RFC 2474: Definition of the DS Field in IPv4 and IPv6 Headers
- RFC 2597: Assured Forwarding PHB Group
- RFC 3246: Expedited Forwarding PHB
- Cisco QoS Design Guides